

The background of the slide is a detailed painting of the Tower of Babel. The tower is a massive, multi-tiered structure made of brick and stone, with numerous arches and windows. It rises steeply from a riverbank, with a city and harbor at its base. In the foreground, a group of people in historical attire are gathered, some looking up at the tower. The sky is filled with soft, white clouds.

Machine Translation: Challenges and Approaches

Based on a talk by **Nizar Habash**

<http://www.nizarhabash.com/>

Professor of Computer Science,
New York University Abu Dhabi



Google offers translations between many languages: Afrikaans
Albanian Amharic history Arabic Armenian Azerbaijani Basque
Belarusian Bengali Bosnian Bulgarian Catalan Cebuano
Chichewa Chinese Corsican Croatian Czech Danish history
Dutch check English Esperanto Estonian Filipino Finnish
history French Frisian Galician Georgian history German
Greek Gujarati Haitian Creole Hausa Hawaiian Hebrew Hindi
Hmong Hungarian Icelandic Igbo Indonesian Irish Italian
Japanese Javanese Kannada Kazakh Khmer Kinyarwanda
Korean Kurdish (Kurmanji) Kyrgyz Lao Latin Latvian
Lithuanian Luxembourgish Macedonian Malagasy Malay
Malayalam Maltese Maori Marathi Mongolian Myanmar
(Burmese) Nepali Norwegian Odia (Oriya) Pashto Persian
Polish Portuguese Punjabi Romanian Russian Samoan Scots
Gaelic Serbian Sesotho Shona Sindhi Sinhala Slovak
Slovenian Somali Spanish Sundanese Swahili Swedish Tajik
Tamil Tatar Telugu Thai Turkish Turkmen Ukrainian Urdu
Uyghur Uzbek Vietnamese Welsh Xhosa Yiddish Yoruba Zulu

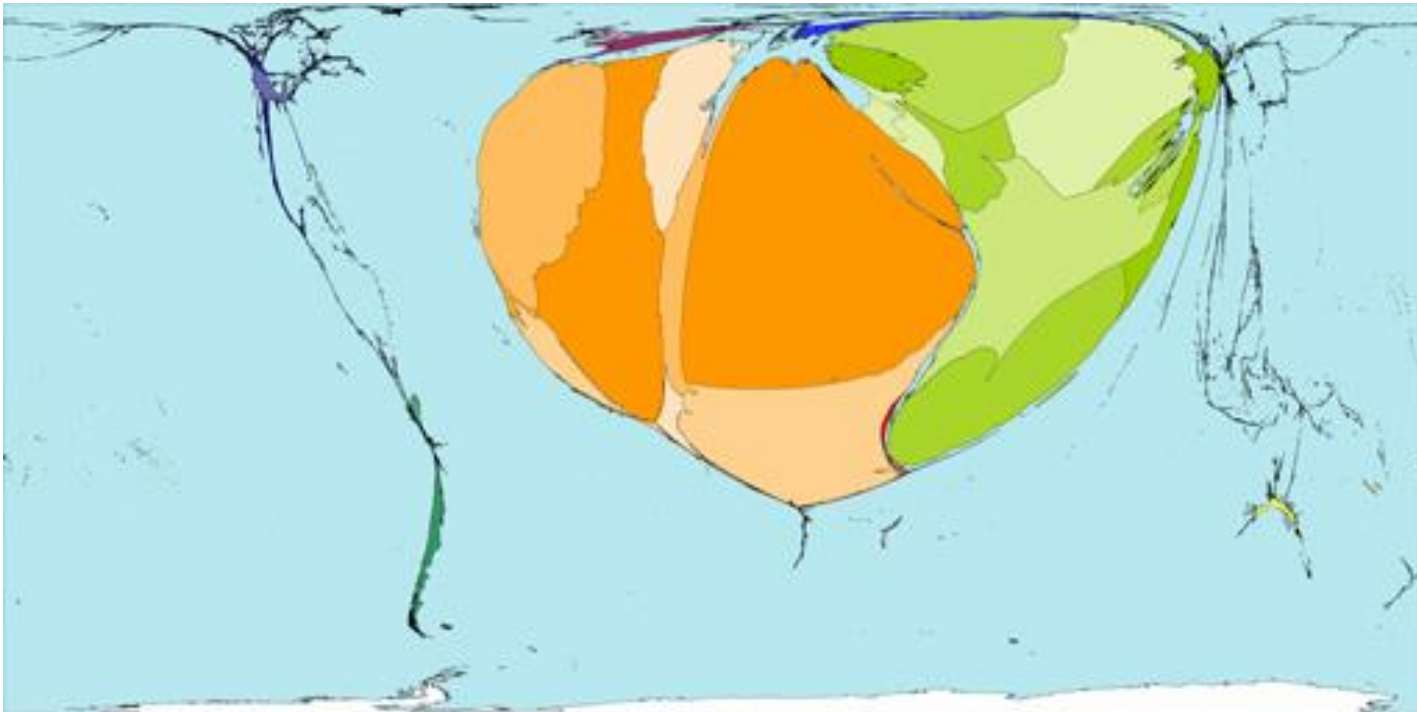
WorldMapper

The world as you've never seen it before

List of languages with maps showing world-wide distribution:

http://www.worldmapper.org/extraindex/text_language.html

Eg Arabic:



Road Map

- Multilingual Challenges for MT
- MT Approaches
- MT Evaluation

Multilingual Challenges

- Complex Orthography (writing system)
 - Ambiguous spelling, eg Arabic vowels omitted
 - كَتَبَ الأولادُ اشْعَاراً كتب الاولاد اشعارا
 - Ambiguous word boundaries, eg Chinese
 - 美单方削减中国纺织品出口配额
- Lexical Ambiguity
 - **Bank** → بنك (financial) vs. ضفة (river)
 - **Eat** → essen (human) vs. fressen (animal)

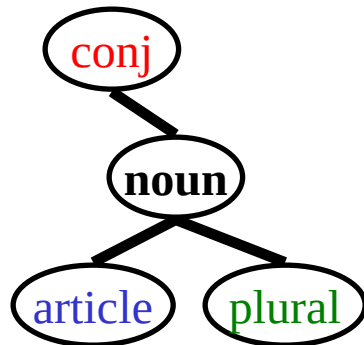
Multilingual Challenges

Morphological complexity and variation

- Affixation vs. Root+Pattern

write	→	written	كتب	→	مكتوب
kill	→	killed	قتل	→	مقتول
do	→	done	فعل	→	مفعول

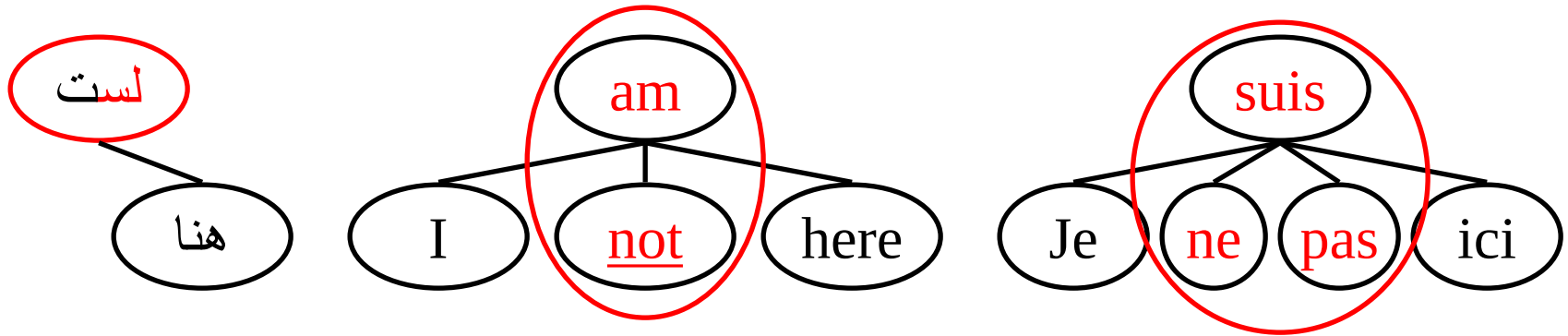
- Tokenization: a “word” can be a whole phrase



<i>And the cars</i>	→	<i>and the cars</i>
والسيارات	→	w Al SyArAt
Et les voitures	→	et les voitures

Translation Divergences

conflation



أست هنا

I-am-not here

I am not here

Je ne suis pas ici

I not am not here

Translation Divergences

head swap and categorial

English	John swam across the river quickly
Spanish	Juan cruzó rápidamente el río nadando <i>Gloss: John crossed fast the river swimming</i>
Arabic	اسرع جون عبور النهر سباحة <i>Gloss: sped john crossing the-river swimming</i>
Chinese	约翰 快速 地 游 过 这 条 河 <i>Gloss: John quickly (DE) swam cross the (Quantifier) river</i>
Russian	Джон быстро переплыл реку <i>Gloss: John quickly cross-swam river</i>

Corpus resources for training

Need a Corpus (eg web-as-corpus: WebBootCat) and DICTIONARY, other NLP resources.

BUT: really need PARALLEL corpus, with SOURCE and TARGET sentences ALIGNED.

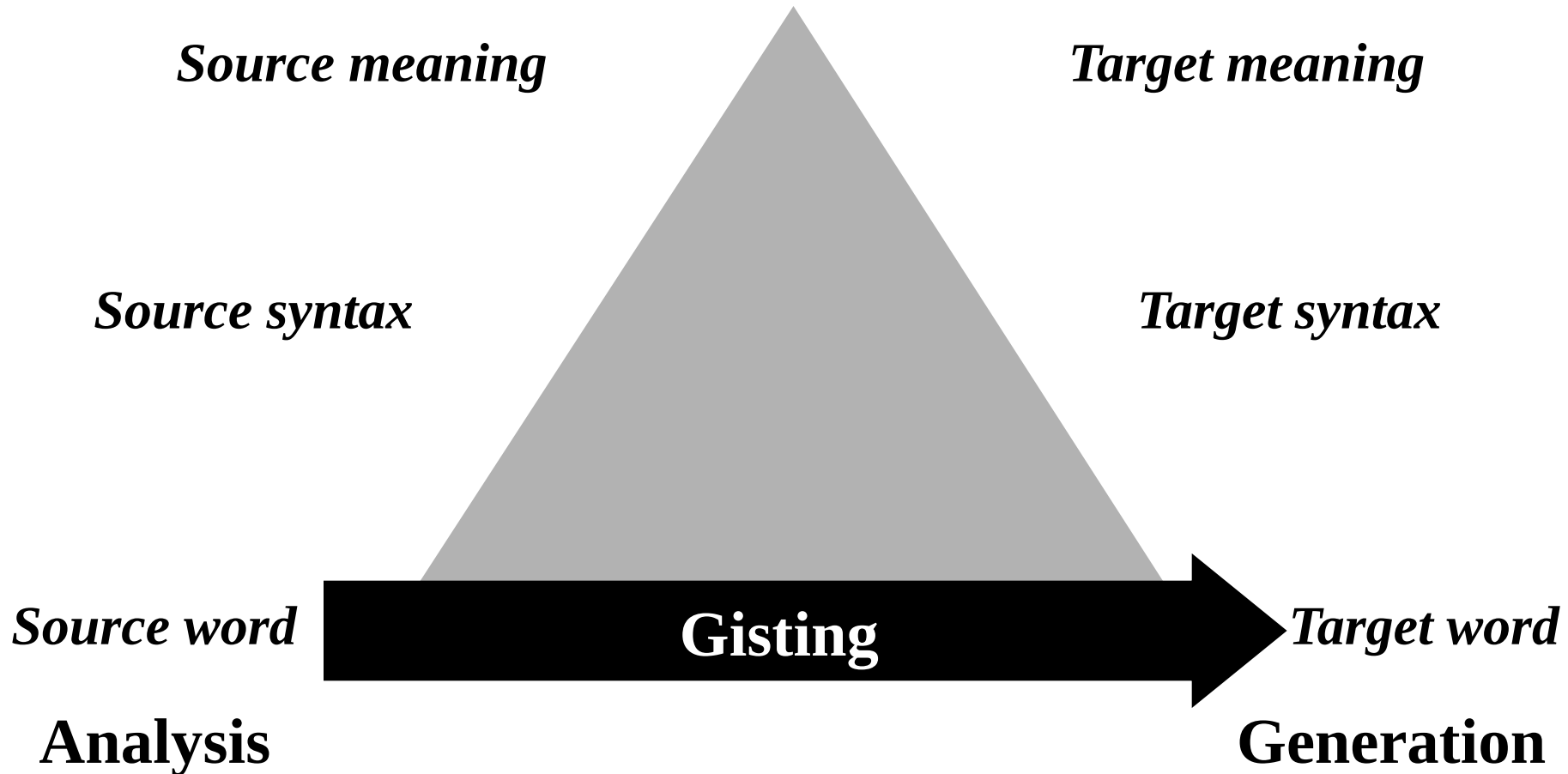
Some languages have few resources, esp non-European languages: Bengali, Amharic, ...

Road Map

- Multilingual Challenges for MT
- **MT Approaches**
- MT Evaluation

MT Approaches

MT Pyramid

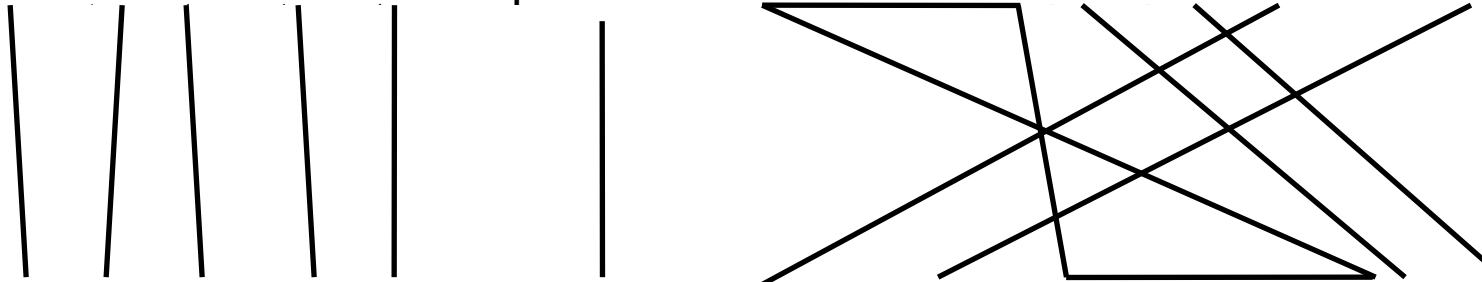


MT Approaches

Gisting Example

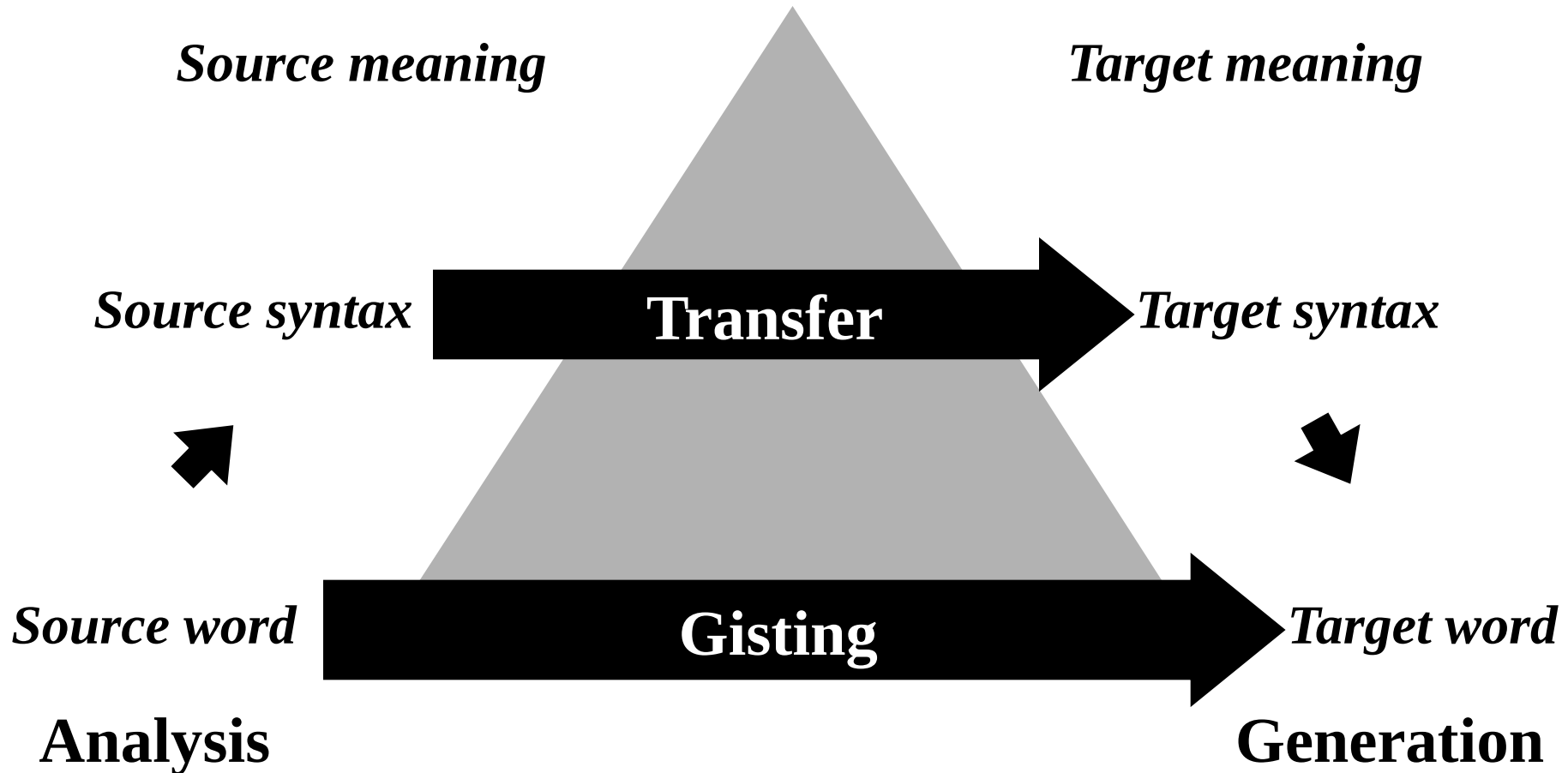
Sobre la base de dichas experiencias se estableció en 1988 una metodología.

On the basis of these experiences, a methodology was arrived at in 1988.



MT Approaches

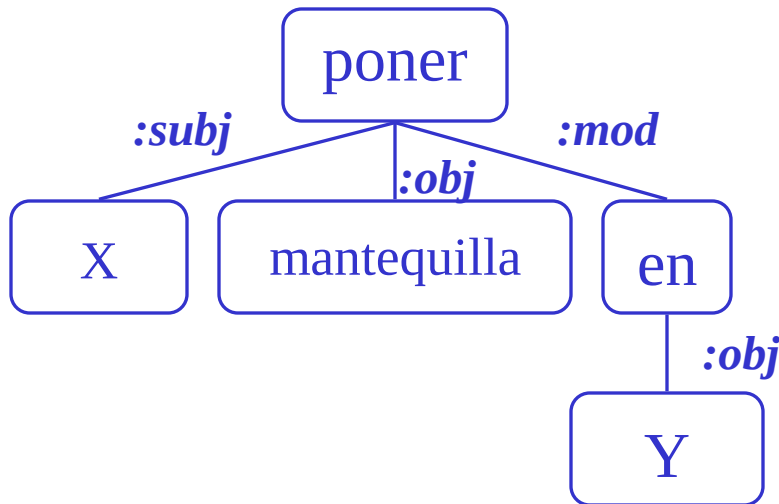
MT Pyramid



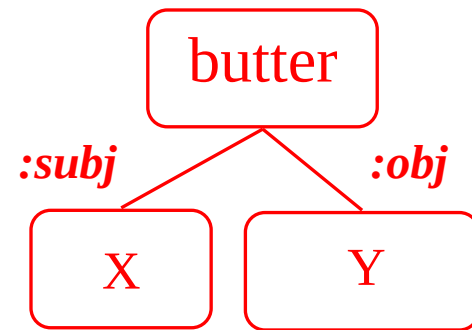
MT Approaches

Transfer Example

- Transfer Lexicon
 - Map SL structure to TL structure



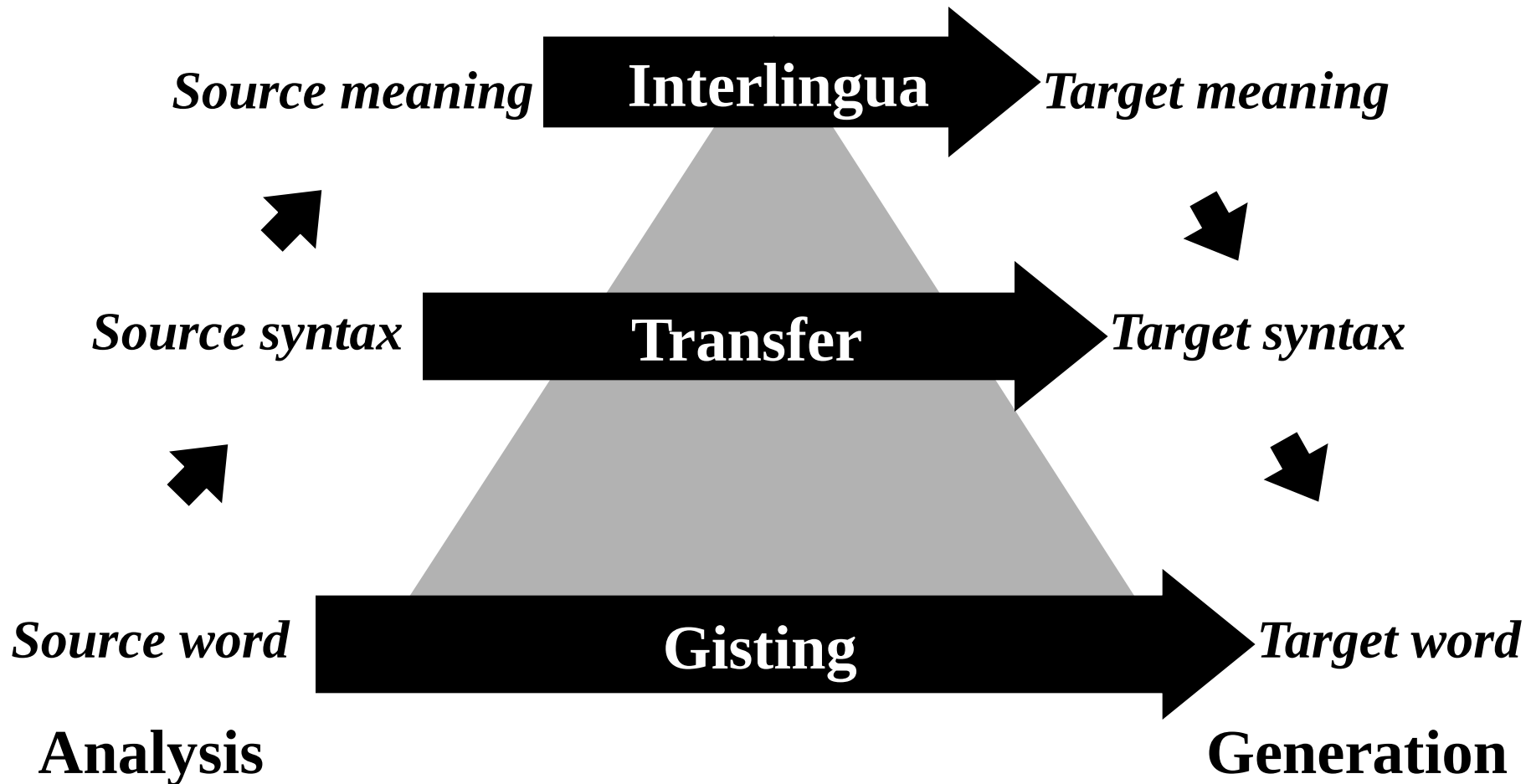
X puso mantequilla en Y



X buttered Y

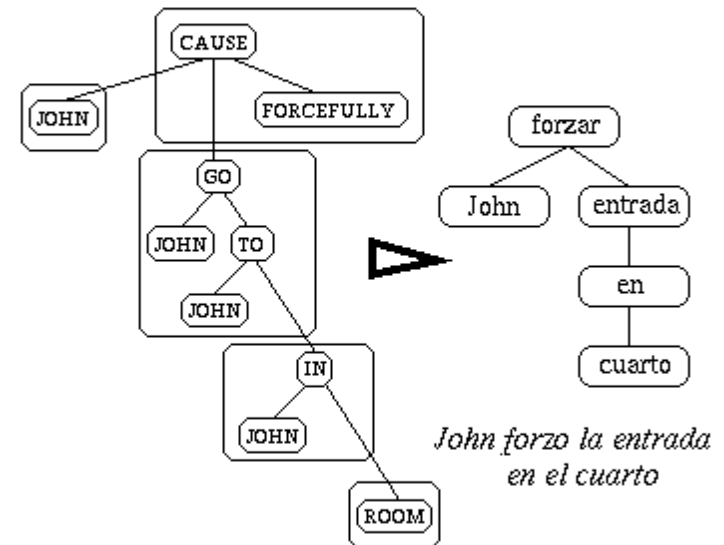
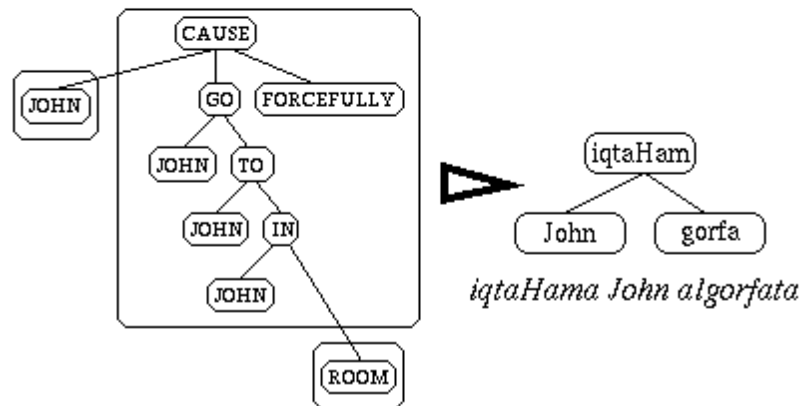
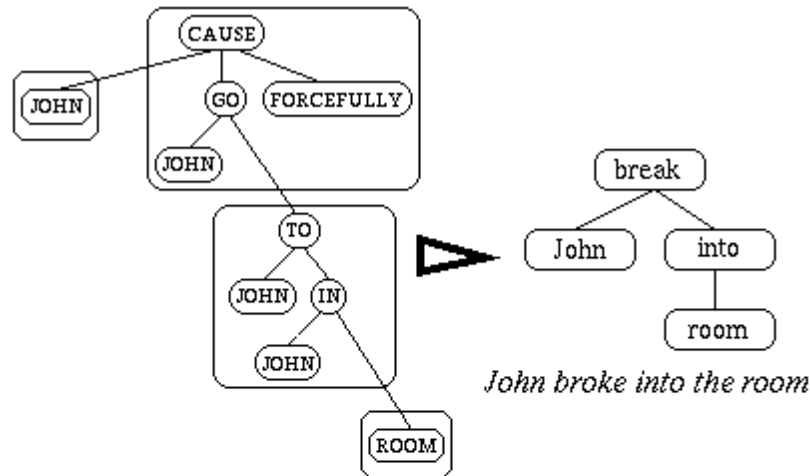
MT Approaches

MT Pyramid



MT Approaches

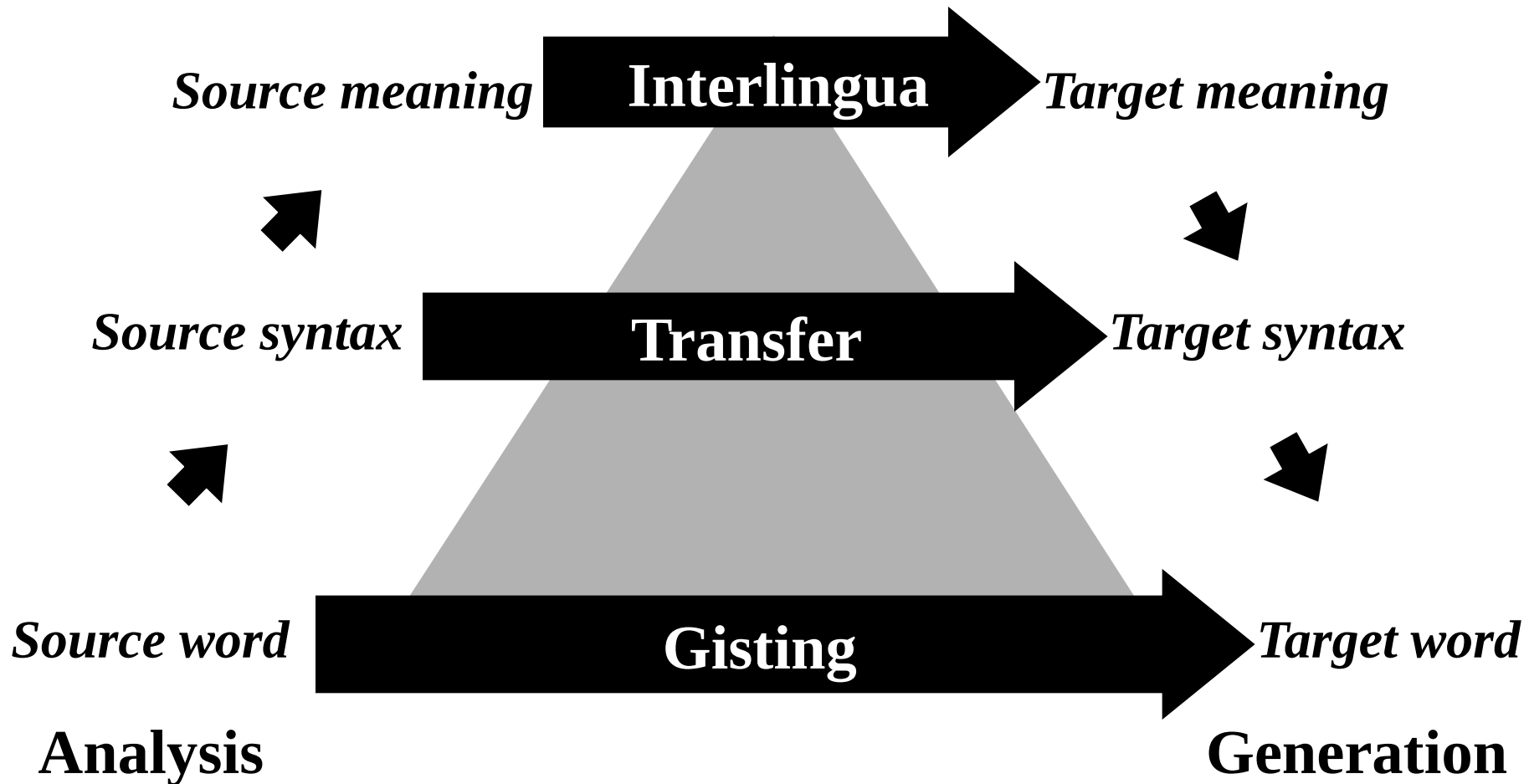
Interlingua Example: Lexical Conceptual Structure



(Dorr, 1993)

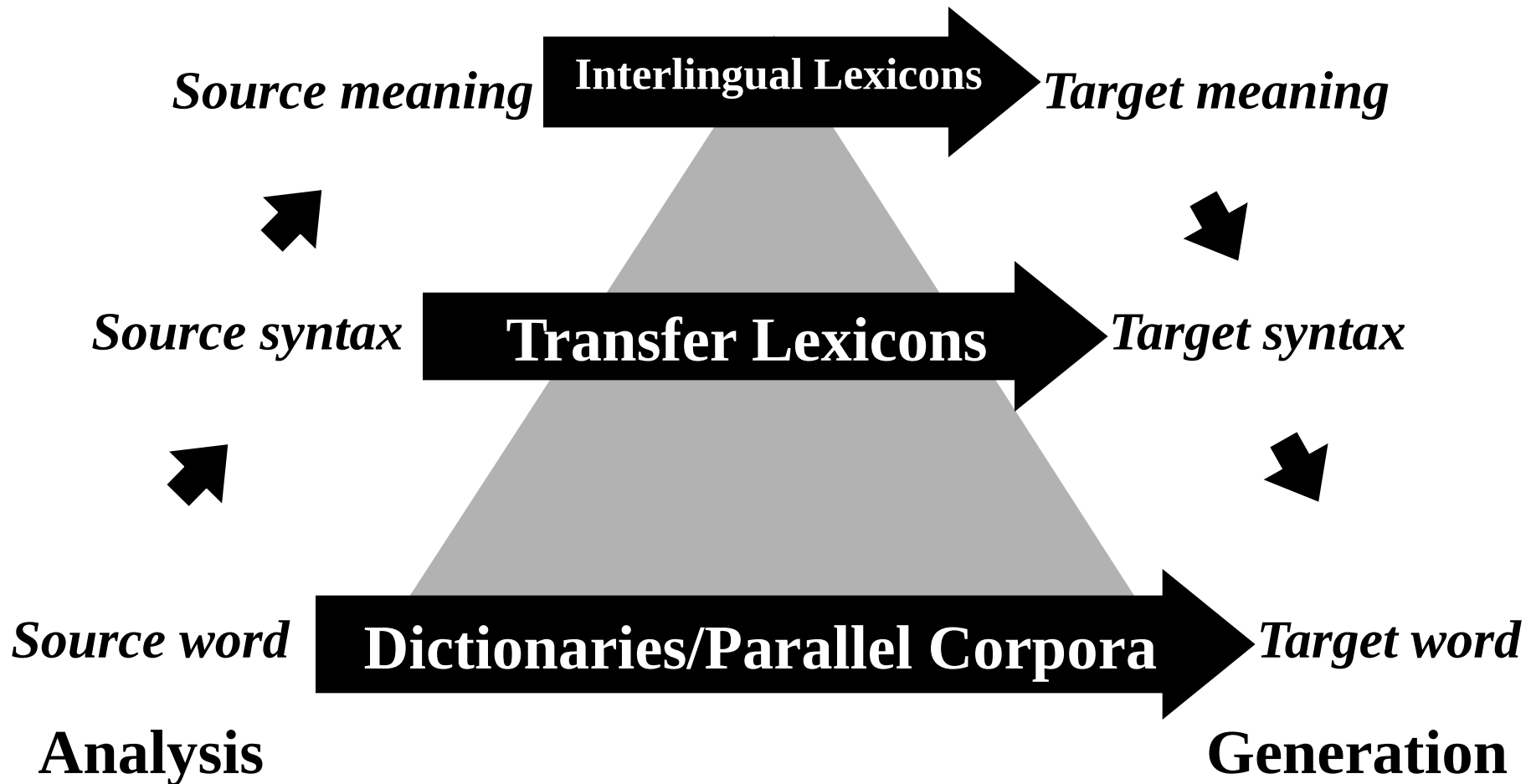
MT Approaches

MT Pyramid



MT Approaches

MT Pyramid



Statistical MT

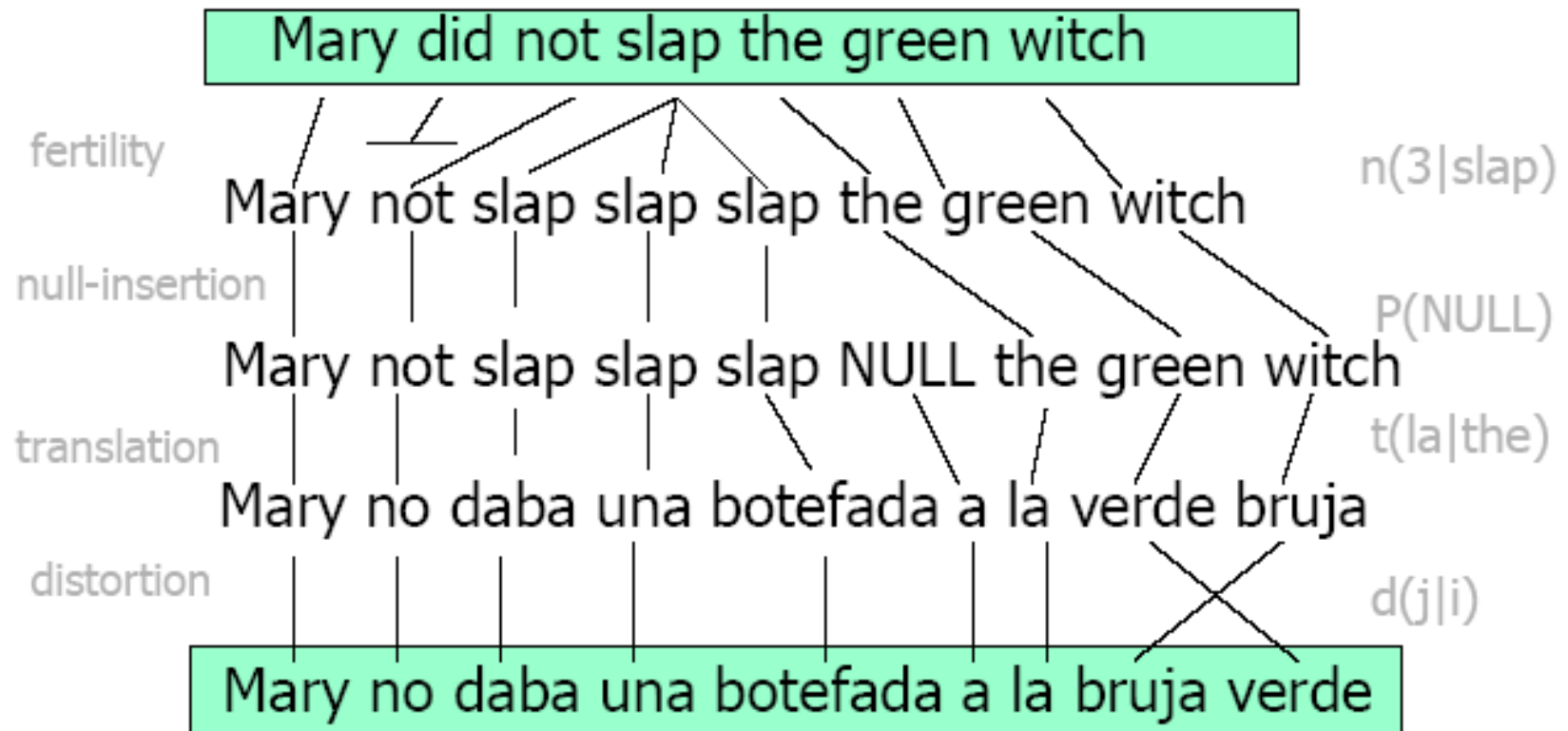
Automatic Word Alignment

- GIZA++
 - A statistical machine translation toolkit used to train word alignments.
 - Uses Expectation-Maximization with various constraints to bootstrap alignments

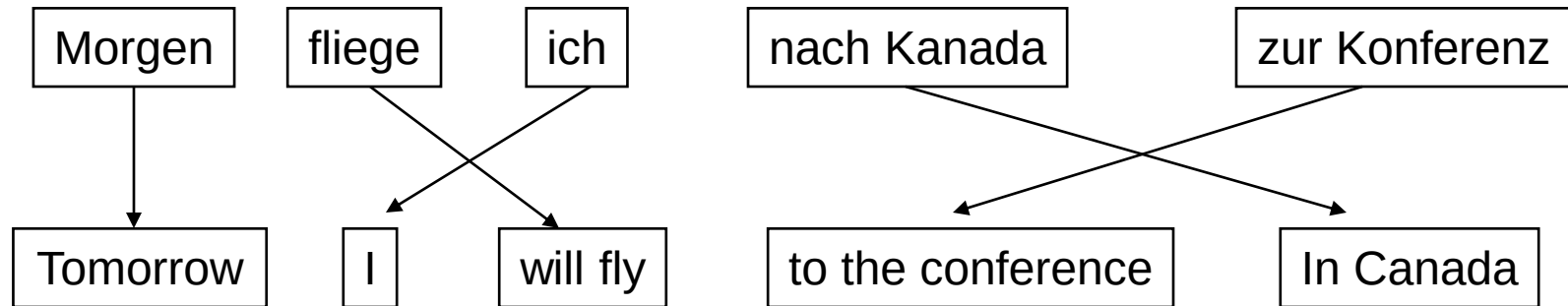
	Maria	no	dio	una	bofetada	a	la	bruja	verde
Mary									
did									
not									
slap									
the									
green									
witch									

Statistical MT

IBM Model (Word-based Model)



Phrase-Based Statistical MT



- Foreign input segmented in to phrases
 - “phrase” is any sequence of words
- Each phrase is probabilistically translated into English
 - $P(\text{to the conference} \mid \text{zur Konferenz})$
 - $P(\text{into the meeting} \mid \text{zur Konferenz})$
- Phrases are probabilistically re-ordered

Word Alignment Induced Phrases

Maria no dió una bofetada a la bruja verde

Mary								
did								
not								
slap								
the								
green								
witch								

(Maria, Mary) (no, did not) (slap, dió una bofetada) (la, the) (bruja, witch) (verde, green)

Word Alignment Induced Phrases

	Maria	no	dió	una	bofetada	a	la	bruja	verde
Mary									
did									
not									
slap									
the									
green									
witch									

(Maria, Mary) (no, did not) (slap, dió una bofetada) (la, the) (bruja, witch) (verde, green)
(a la, the) (dió una bofetada a, slap the)

Word Alignment Induced Phrases

Maria no dió una bofetada a la bruja verde

Mary	■	□	□	□	□	□	□	□
did	□	■	□	□	□	□	□	□
not	□	■	□	□	□	□	□	□
slap	□	□	■	■	■	□	□	□
the	□	□	□	□	□	■	□	□
green	□	□	□	□	□	□	□	■
witch	□	□	□	□	□	□	■	□

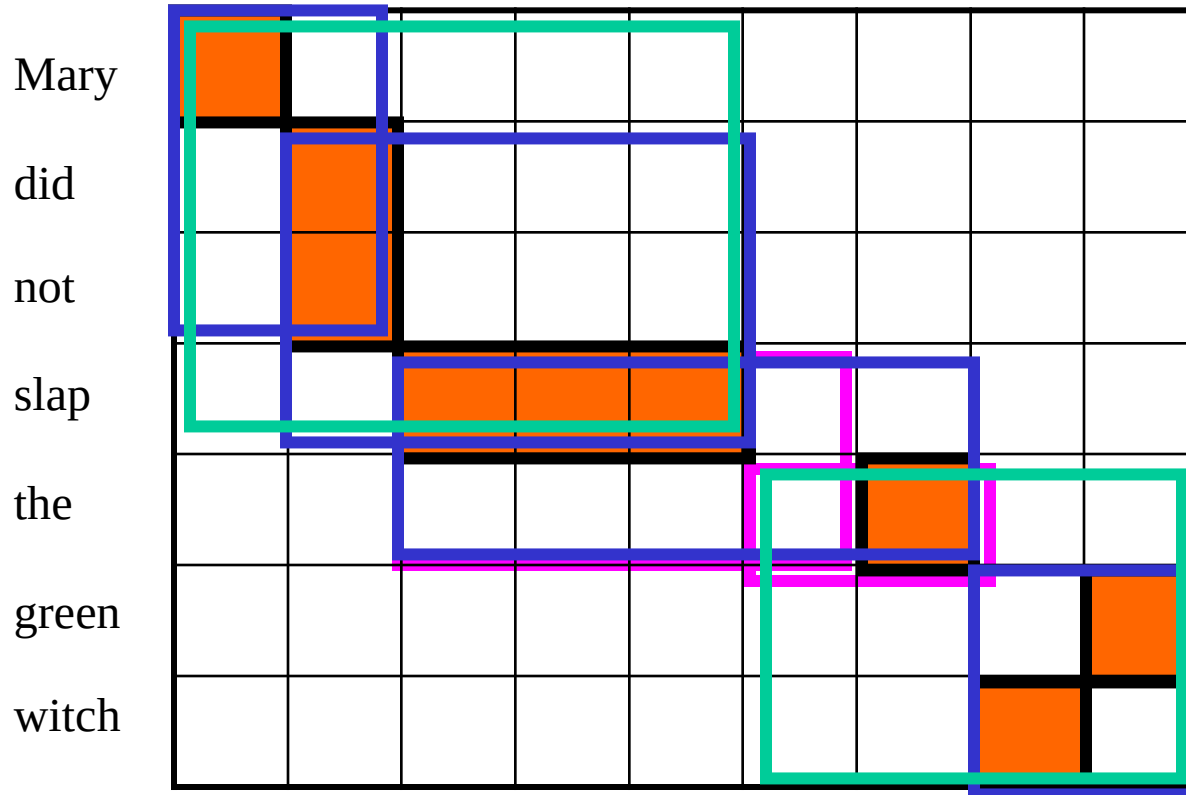
(Maria, Mary) (no, did not) (slap, dió una bofetada) (la, the) (bruja, witch) (verde, green)

(a la, the) (dió una bofetada a, slap the)

(Maria no, Mary did not) (no dió una bofetada, did not slap), (dió una bofetada a la, slap the)
(bruja verde, green witch)

Word Alignment Induced Phrases

Maria no dió una bofetada a la bruja verde



(Maria, Mary) (no, did not) (slap, dió una bofetada) (la, the) (bruja, witch) (verde, green)

(a la, the) (dió una bofetada a, slap the)

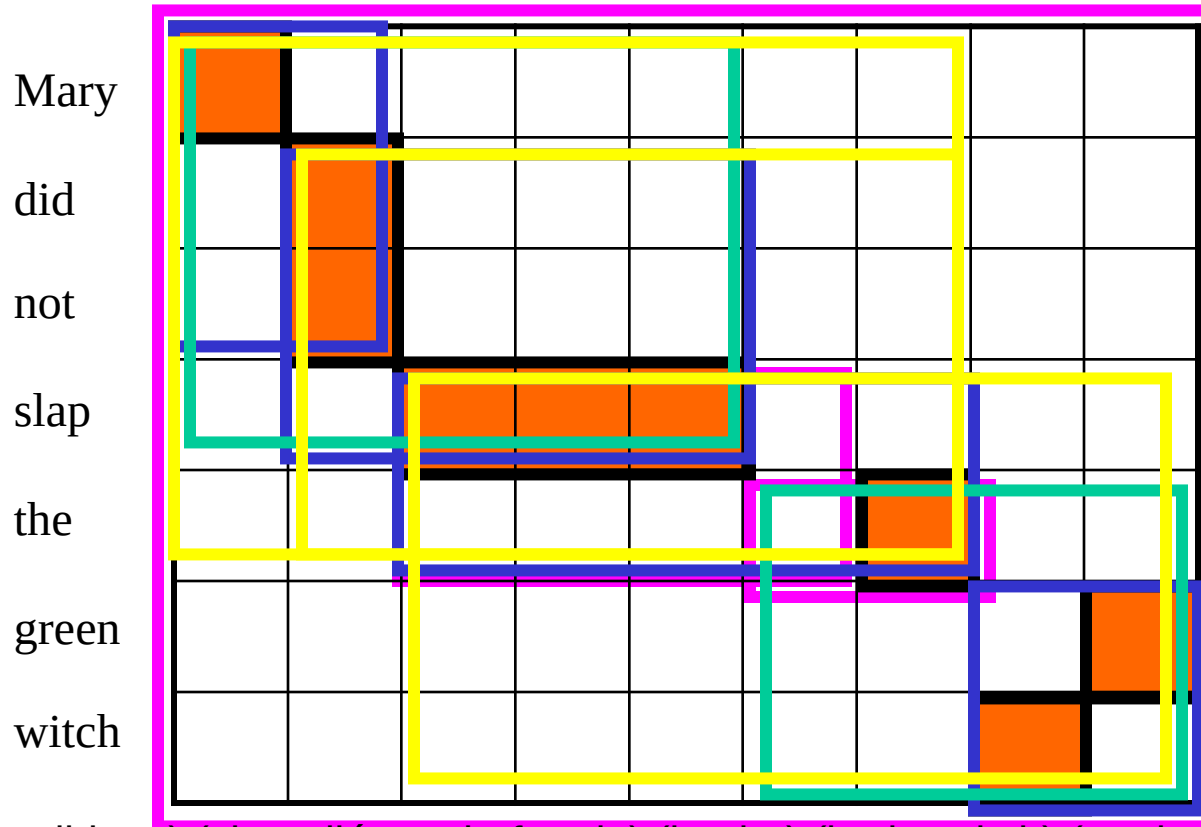
(Maria no, Mary did not) (no dió una bofetada, did not slap), (dió una bofetada a la, slap the)

(bruja verde, green witch) (Maria no dió una bofetada, Mary did not slap)

(a la bruja verde, the green witch) ...

Word Alignment Induced Phrases

Maria no dió una bofetada a la bruja verde



(Maria, Mary) (no, did not) (slap, dió una bofetada) (la, the) (bruja, witch) (verde, green)

(a la, the) (dió una bofetada a, slap the)

(Maria no, Mary did not) (no dió una bofetada, did not slap), (dió una bofetada a la, slap the)

(bruja verde, green witch) (Maria no dió una bofetada, Mary did not slap)

(a la bruja verde, the green witch) ...

(Maria no dió una bofetada a la bruja verde, Mary did not slap the green witch)

Advantages of Phrase-Based SMT

- Many-to-many mappings can handle non-compositional phrases
- Local context is very useful for disambiguating
 - “Interest rate” → ...
 - “Interest in” → ...
- The more data, the longer the learned phrases
 - Sometimes whole sentences

Road Map

- Multilingual Challenges for MT
- MT Approaches
- MT Evaluation

MT Evaluation

- More art than science
- Wide range of Metrics/Techniques
 - interface, ..., scalability, ..., faithfulness, ...
space/time complexity, ... etc.
- Automatic vs. Human-based
 - *Dumb Machines vs. Slow Humans*

Human-based Evaluation Example

Fidelity / Accuracy Criteria

5	contents of original sentence conveyed (might need minor corrections)
4	contents of original sentence conveyed BUT errors in word order
3	contents of original sentence generally conveyed BUT errors in relationship between phrases, tense, singular/plural, etc.
2	contents of original sentence not adequately conveyed, portions of original sentence incorrectly translated, missing modifiers
1	contents of original sentence not conveyed, missing verbs, subjects, objects, phrases or clauses

Human-based Evaluation Example

Fluency / Intelligibility Criteria

5	clear meaning, good grammar, terminology and sentence structure
4	clear meaning BUT bad grammar, bad terminology or bad sentence structure
3	meaning graspable BUT ambiguities due to bad grammar, bad terminology or bad sentence structure
2	meaning unclear BUT inferable
1	meaning absolutely unclear

Semi-Automatic Evaluation Example

Bleu Metric

(Papineni et al 2001)

- Bleu
 - *BiLingual Evaluation Understudy*
 - Modified n-gram precision with length penalty
 - Quick, inexpensive and language independent
 - Correlates highly with human evaluation
 - Compares MT output against several human translations to give a standardized score

Automatic Evaluation Example

Bleu Metric

Test Sentence
(MT output)

colorless green ideas sleep furiously

Gold Standard References
(human translations)

all dull jade ideas sleep irately
drab emerald concepts sleep furiously
colorless immature thoughts nap angrily

Automatic Evaluation Example

Bleu Metric

Test Sentence

colorless green ideas sleep furiously

Gold Standard References

all dull jade ideas sleep irately
drab emerald concepts sleep furiously
colorless immature thoughts nap angrily

Unigram precision = 4/5

Automatic Evaluation Example

Bleu Metric

Test Sentence

Gold Standard References

colorless green

green ideas

ideas sleep

sleep furiously

all dull jade ideas sleep irately

drab emerald concepts sleep furiously

colorless immature thoughts nap angrily

Unigram precision = $4 / 5 = 0.8$

Bigram precision = $2 / 4 = 0.5$

Bleu Score = $(a_1 a_2 \dots a_n)^{1/n}$
 $= (0.8 \times 0.5)^{1/2} = 0.6325 \rightarrow 63.25$

Summary

- Multilingual Challenges for MT:
 - Different writing systems, morphology, segmentation, word order;
 - Need parallel corpus, dictionaries, NLP tools
- MT Approaches: statistical v rule-based
 - Gisting: word-for-word;
 - Transfer: source-to-target phrase mapping;
 - Interlingua: map to/from “semantic representation”
- MT Evaluation: Accuracy, Fluency
 - IBM BLEU method:
count overlaps with Reference (human) translations